جامعة أم القرى
UMM AL-QURA UNIVERSITY

Course: Natural Language Processing (NLP)

Sentiment Classification System Using Transformer Models (BERT, RoBERTa, and DistilBERT)

Submitted by:

Ghala Sulaiman

Student ID: 444007681

Asael Ibrahim Alharbi

Student ID: 444011109

Aseel Ahmad Alsubhi

Student ID: 444000725

Instructor:

Dr. Mashael-Allihaybi

Date of Submission: 21/5/2026

1. Introduction

Natural Language Processing (NLP) is one of the most important fields in Artificial Intelligence. It focuses on enabling computers to understand, process, and analyze human language in a meaningful way. NLP is widely used in many applications such as chatbots, translation systems, speech recognition, and sentiment analysis.

Sentiment Analysis is a popular NLP task that aims to determine the emotional tone of a text, such as whether the opinion expressed is positive or negative. It is commonly used in social media analysis, customer reviews, and opinion mining to better understand user feedback and public sentiment.

In recent years, transformer-based models have achieved significant improvements in NLP tasks. Models such as BERT, RoBERTa, and DistilBERT provide powerful language understanding capabilities and high performance in text classification tasks.

In this project, a sentiment classification system was developed using three transformer models: BERT, RoBERTa, and DistilBERT. The system was implemented using Streamlit to provide an interactive user interface where users can input text and receive sentiment predictions along with confidence scores.

The main goal of this project is to compare the performance of the three models and analyze their effectiveness in sentiment analysis tasks.

2. Project Objectives

The main objectives of this project are:

- Build a sentiment analysis system using transformer models.
- Compare the performance of BERT, RoBERTa, and DistilBERT.
- Predict whether user input text is positive or negative.
- Display confidence scores for predictions.
- Create an interactive web interface using Streamlit.
- Analyze and compare the results of the three models.

3. Technologies and Tools Used

The following tools and technologies were used in this project:

Tool / Technology	Purpose
Python	Main programming language
Streamlit	Building the web interface
Hugging Face Transformers	Using pretrained NLP models
Pandas	Data processing and handling
PyTorch	Deep learning framework
CSV Dataset	Testing and evaluation

4. Models Used

4.1 BERT

BERT (Bidirectional Encoder Representations from Transformers) is a transformer-based model developed by Google. It is designed to understand the context of words in a sentence by reading text in both directions, from left to right and right to left. BERT provides high accuracy in many Natural Language Processing tasks, including sentiment analysis and text classification.

In this project, BERT was used to classify text sentiment into positive or negative categories.

4.2 RoBERTa

RoBERTa (Robustly Optimized BERT Approach) is an improved version of BERT developed by Meta. It was trained using larger datasets and optimized training methods, which helped improve its performance in NLP tasks. RoBERTa is known for achieving strong results in sentiment analysis and language understanding tasks.

In this project, RoBERTa was used to compare its performance with the other transformer models.

4.3 DistilBERT

DistilBERT is a lightweight and faster version of BERT. It was designed to reduce model size and improve processing speed while maintaining most of BERT's performance. DistilBERT is efficient for applications that require faster predictions and lower computational resources.

In this project, DistilBERT was used as a faster alternative for sentiment classification.

5. System Implementation

The sentiment analysis system was implemented using Streamlit to create an interactive and user-friendly web interface. The application allows users to enter text, choose a transformer model, and receive a sentiment prediction result along with a confidence score.

The system includes the following features:

- Text input field for user sentences
- Selection between BERT, RoBERTa, and DistilBERT
- Sentiment prediction functionality
- Display of prediction confidence score
- Comparison between the three models
- Simple and interactive interface design

The models were integrated using pretrained transformer models from the Hugging Face Transformers library.

6. User Interface

The user interface of the system was developed using Streamlit to provide a simple and interactive experience for users. The interface allows users to enter text, select a transformer model, and view the sentiment prediction results directly.

The interface consists of several main sections:

- Text input section
- Model selection section
- Prediction result display
- Confidence score display
- Model comparison section

The design was created to be simple, clear, and easy to use.

6.1 Main Interface

The figure below shows the main interface of the Sentiment Classification System before any input is provided.

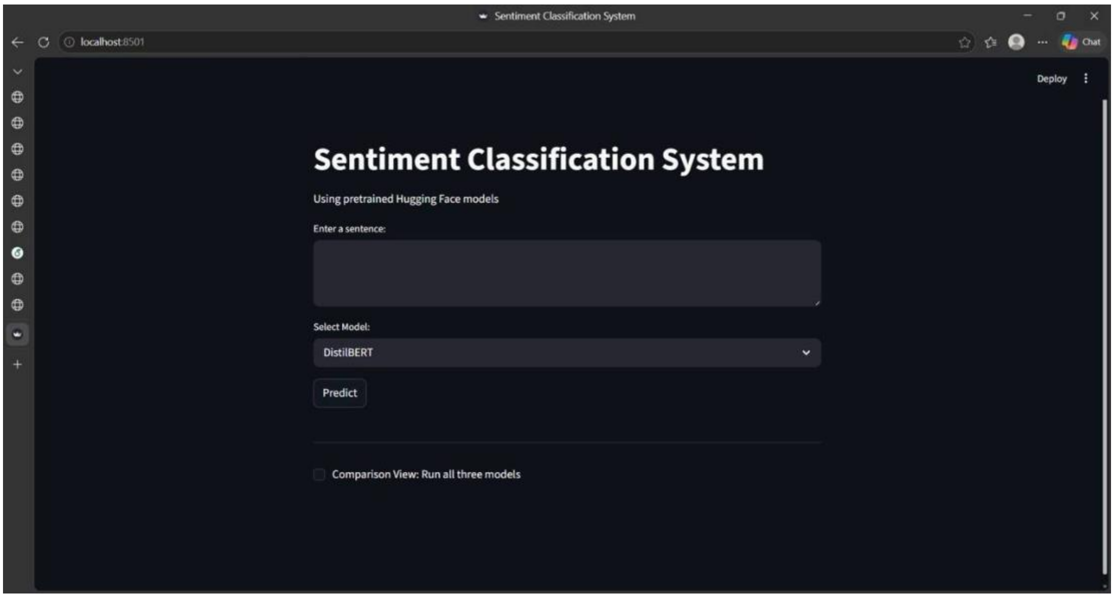

Figure 1: Main interface of the Sentiment Classification System

6.2 Model Selection

Users can select one of three transformer models from the dropdown menu: DistilBERT, RoBERTa, or BERT.

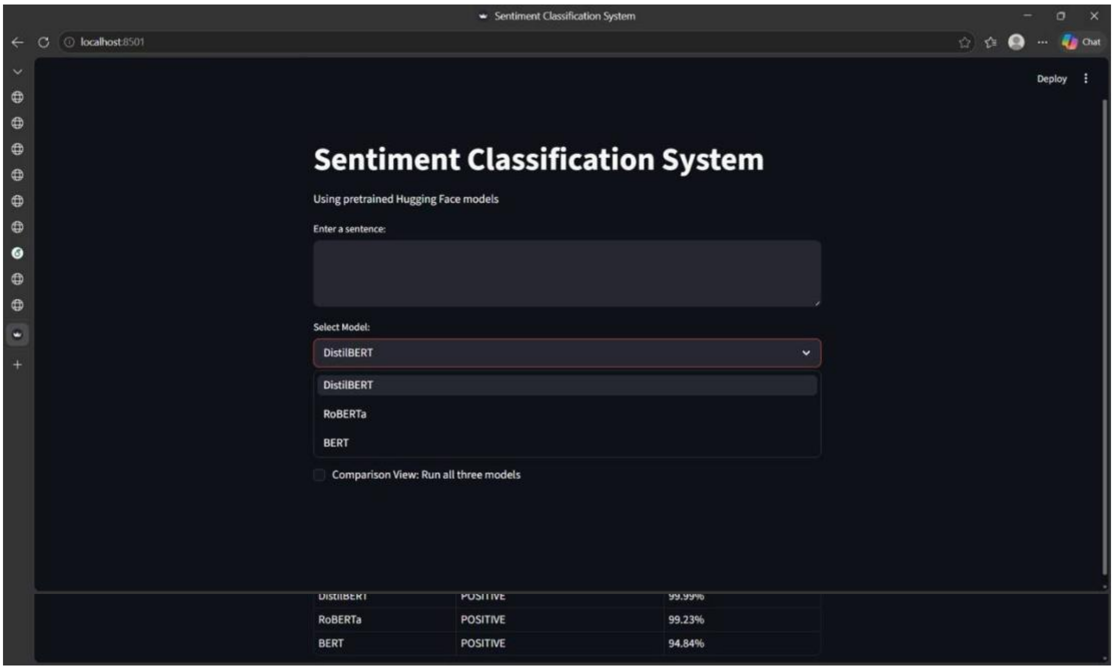

Figure 2: Model selection dropdown showing the three available models.

6.3 Prediction Results

After clicking the Predict button, the system displays the predicted sentiment label along with the confidence score.

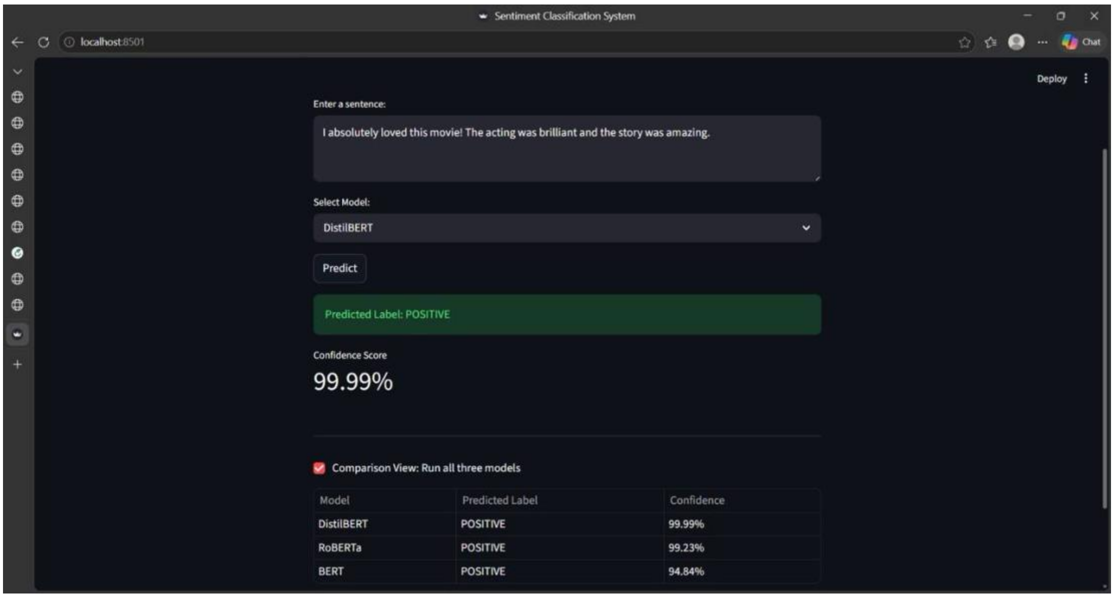

Figure 3: Prediction result displaying the sentiment label and confidence score.

6.4 Model Comparison

The Comparison View runs all three models simultaneously and displays their predictions and confidence scores in a table.

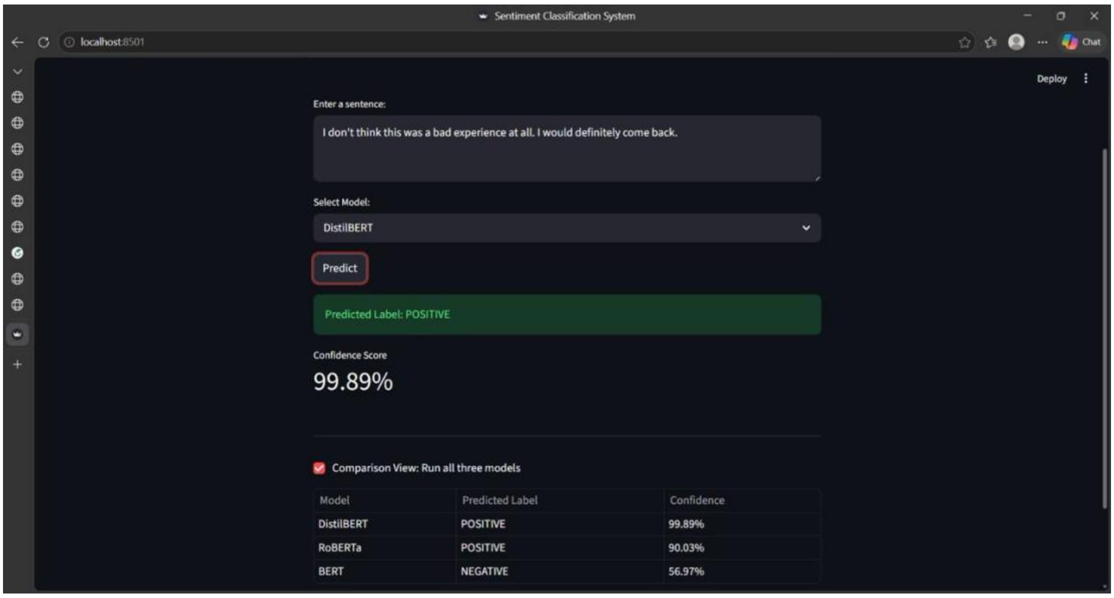

Figure 4: Comparison view showing predictions from all three models simultaneously.

6.5 Confidence Scores

The confidence scores reflect how certain each model is about its prediction, displayed as a percentage.

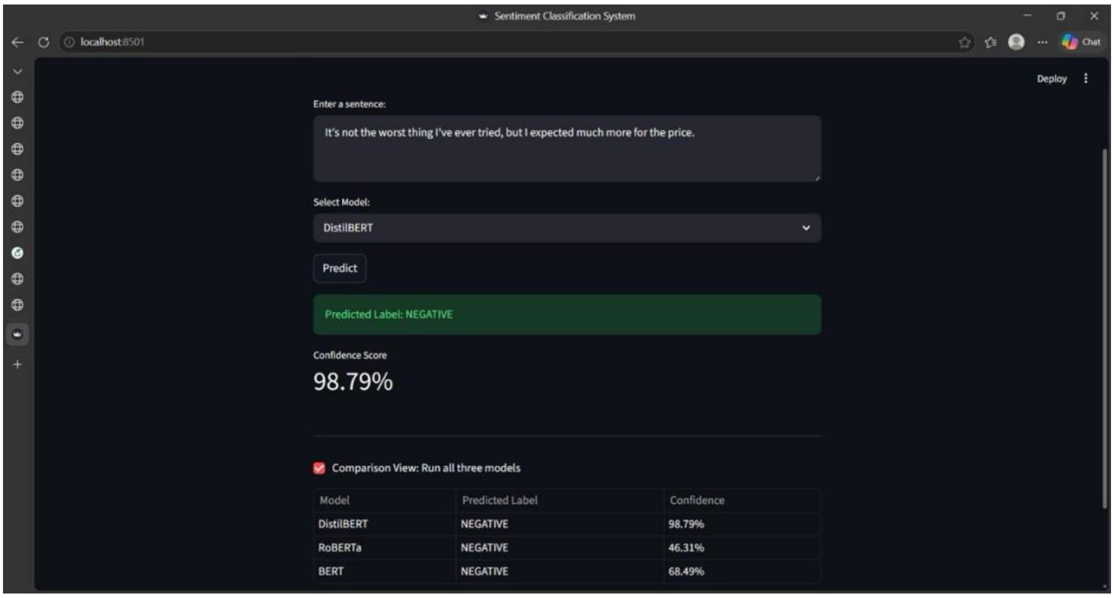

Figure 5: Confidence scores for each model displayed in the comparison view.

7. Dataset

The project used a cleaned dataset stored in a CSV file named clean_dataset.csv. The dataset contains text samples labeled as positive or negative sentiments.

The dataset was used to test and evaluate the performance of the transformer models. Data preprocessing and cleaning were performed before using the dataset in the sentiment classification system.

The dataset helped measure the accuracy and effectiveness of each model during testing and comparison.

8. Experimental Results

This section presents the experimental results obtained from testing the three transformer models: BERT, RoBERTa, and DistilBERT.

The models were evaluated based on several performance metrics, including:

- Accuracy
- Precision
- Recall
- F1-Score

The comparison helps identify which model achieved better sentiment classification performance.

Model	Accuracy	Precision	Recall	F1-Score
BERT	0.51	0.5385	0.14	0.2222
RoBERTa	0.50	0.5000	0.12	0,1935
DistilBERT	0.51	0.5333	0.16	0.2462

Result Analysis

The experimental results show that all three models achieved similar accuracy scores around 0.50–0.51. DistilBERT achieved the highest F1-Score (0.2462), followed by BERT (0.2222), and RoBERTa (0.1935). The low Recall values across all models suggest that the dataset characteristics affected the prediction balance. Overall, DistilBERT demonstrated the most consistent performance among the three models.

9. Testing Sentences

To evaluate the performance of the transformer models, several testing sentences were used. Each sentence was analyzed using BERT, RoBERTa, and DistilBERT to compare prediction results and confidence scores.

The testing process helped measure the consistency and effectiveness of each model in sentiment classification tasks.

Sentence	BERT	RoBERTa	DistilBERT
Sentence 1	POSITIVE	POSITIVE	POSITIVE
Sentence 2	NEGATIVE	NEGATIVE	NEGATIVE
Sentence 3	NEGATIVE	NEGATIVE	NEGATIVE
Sentence 4	NEGATIVE	POSITIVE	POSITIVE
Sentence 5	NEGATIVE	NEGATIVE	NEGATIVE

Sentence	Model	Prediction	Confidence Score
Sentence 1	BERT	POSITIVE	94.84%
Sentence 1	RoBERTa	POSITIVE	99.23%
Sentence 1	DistilBERT	POSITIVE	99.99%

Sentence 2	BERT	NEGATIVE	96.45%
Sentence 2	RoBERTa	NEGATIVE	98.35%
Sentence 2	DistilBERT	NEGATIVE	99.98%
Sentence 3	BERT	NEGATIVE	52.16%
Sentence 3	RoBERTa	NEGATIVE	76.20%
Sentence 3	DistilBERT	NEGATIVE	99.88%
Sentence 4	BERT	NEGATIVE	56.97%
Sentence 4	RoBERTa	POSITIVE	90.62%
Sentence 4	DistilBERT	POSITIVE	99.89%
Sentence 5	BERT	NEGATIVE	68.49%
Sentence 5	RoBERTa	NEGATIVE	46.20%
Sentence 5	DistilBERT	NEGATIVE	98.79%

The following screenshots show the system output for each test sentence using the Comparison View.

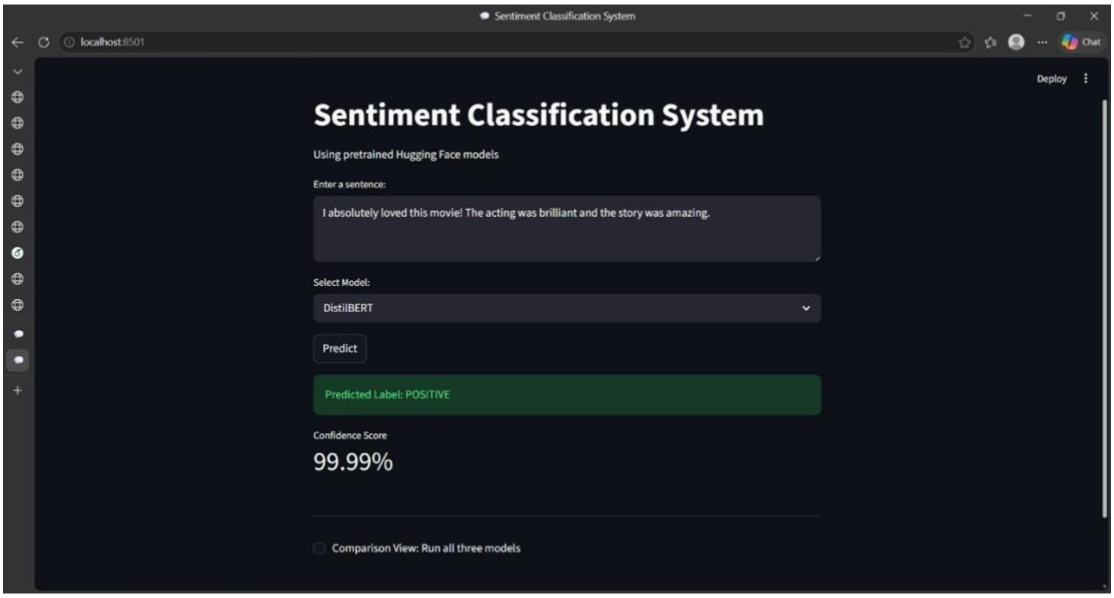

Figure 6: Sentence 1 tested on DistilBERT — Predicted POSITIVE with 99.99% confidence.

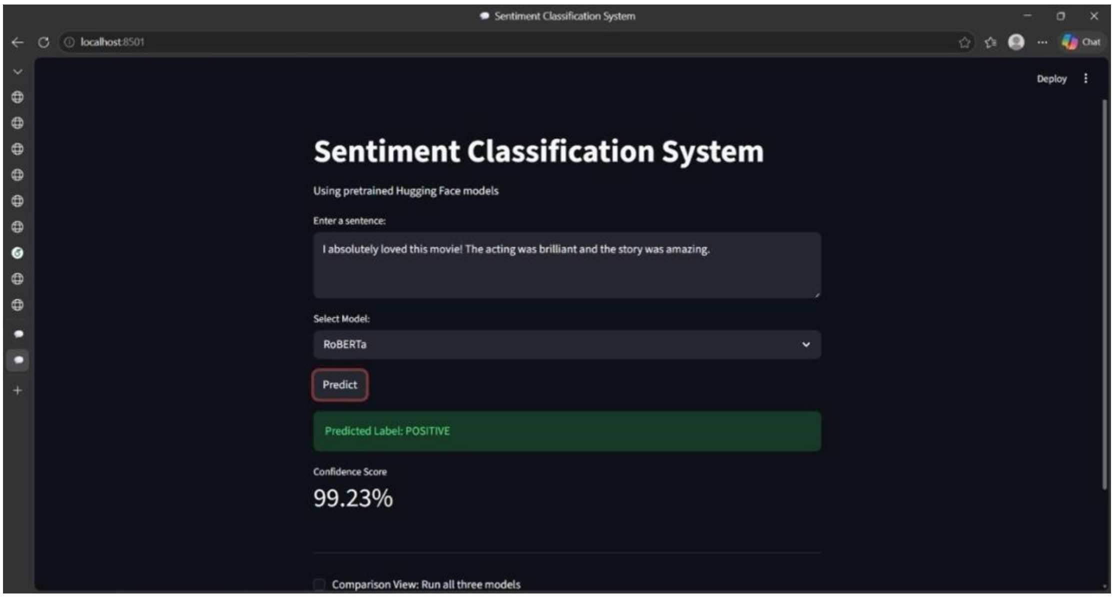

Figure 7: Sentence 1 tested on RoBERTa — Predicted POSITIVE with 99.23% confidence.

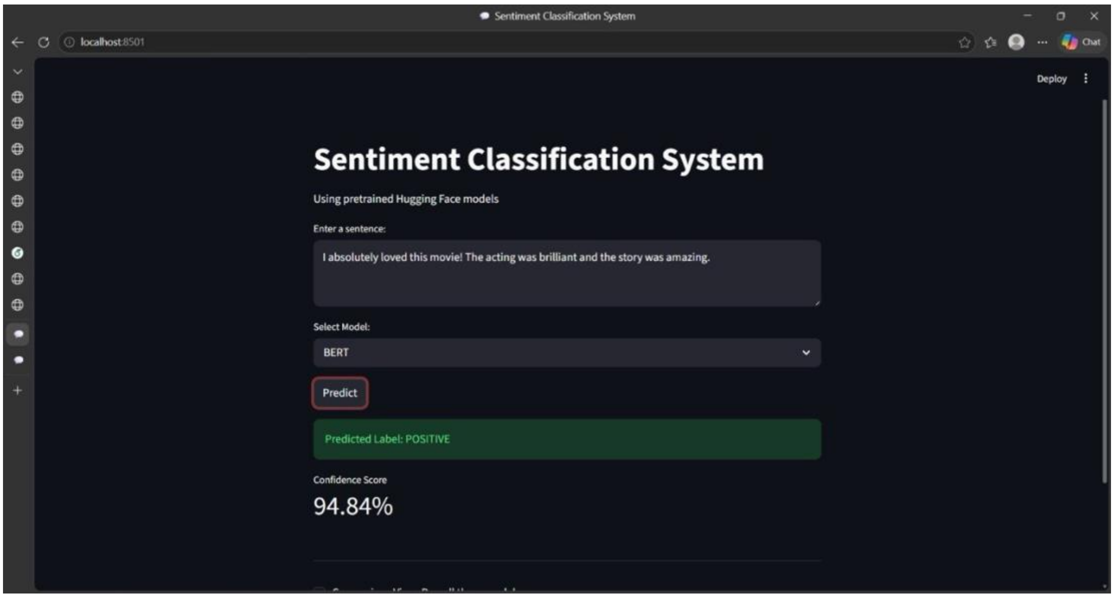

Figure 8: Sentence 1 tested on BERT — Predicted POSITIVE with 94.84% confidence.

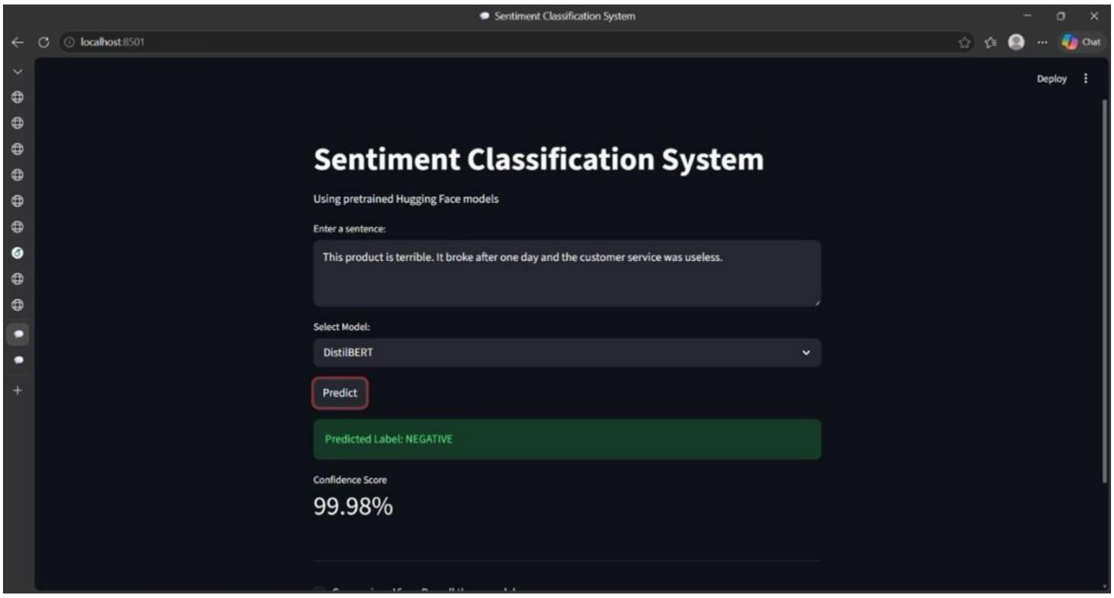

Figure 9: Sentence 2 tested on DistilBERT — Predicted NEGATIVE with 99.98% confidence..

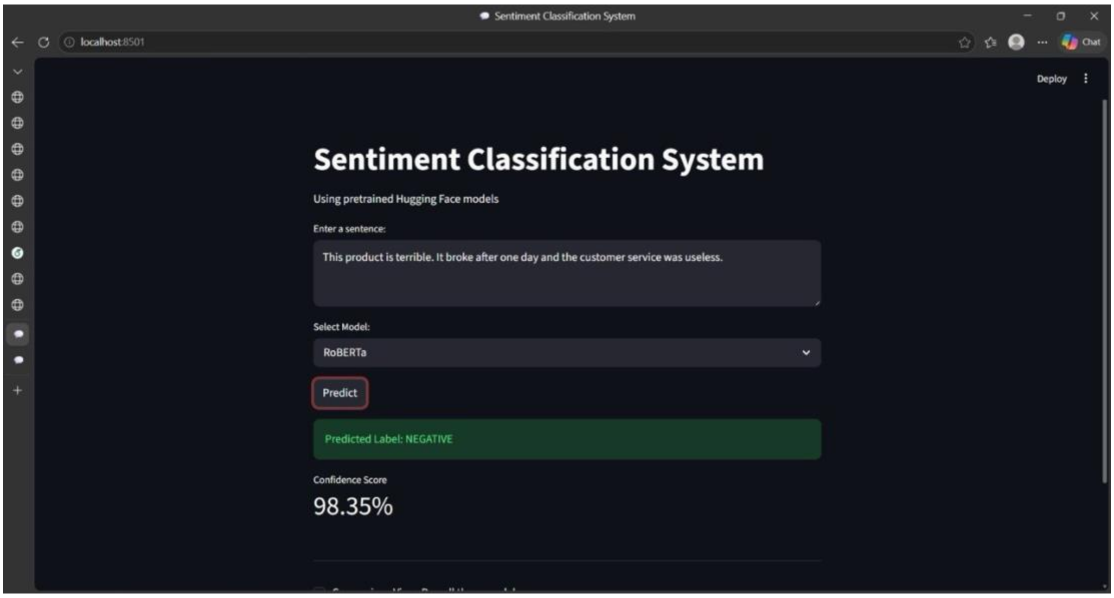

Figure 10: Sentence 2 tested on RoBERTa — Predicted NEGATIVE with 98.35% confidence.

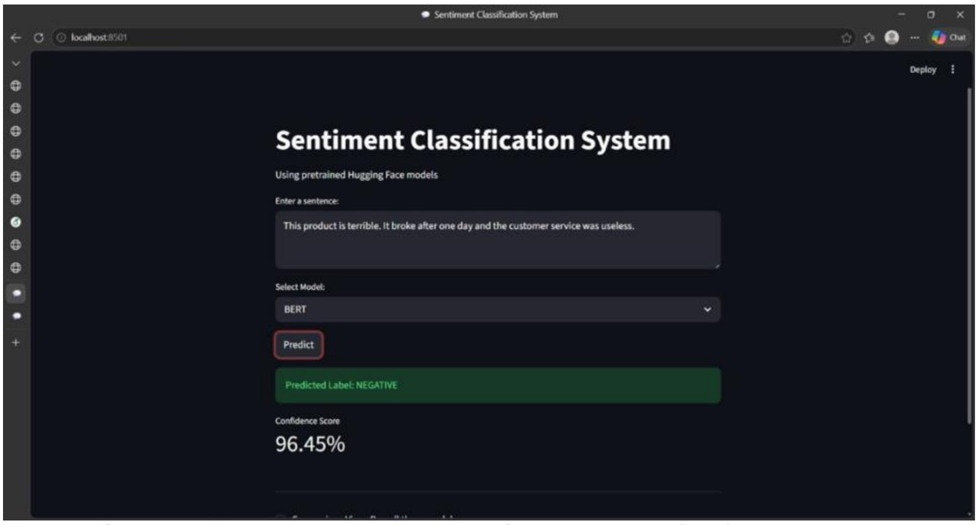

Figure 11: Sentence 2 tested on BERT — Predicted NEGATIVE with 96.45% confidence.

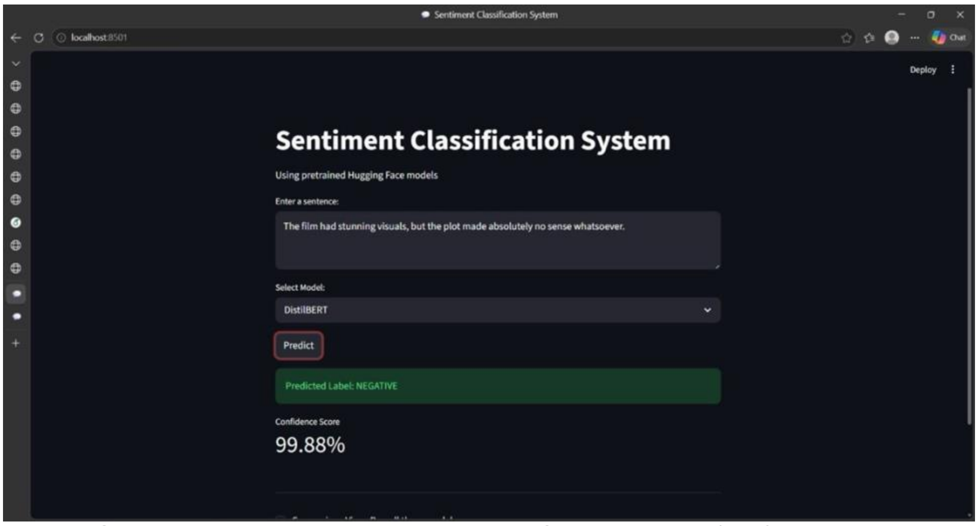

Figure 12: Sentence 3 tested on DistilBERT — Predicted NEGATIVE with 99.88% confidence.

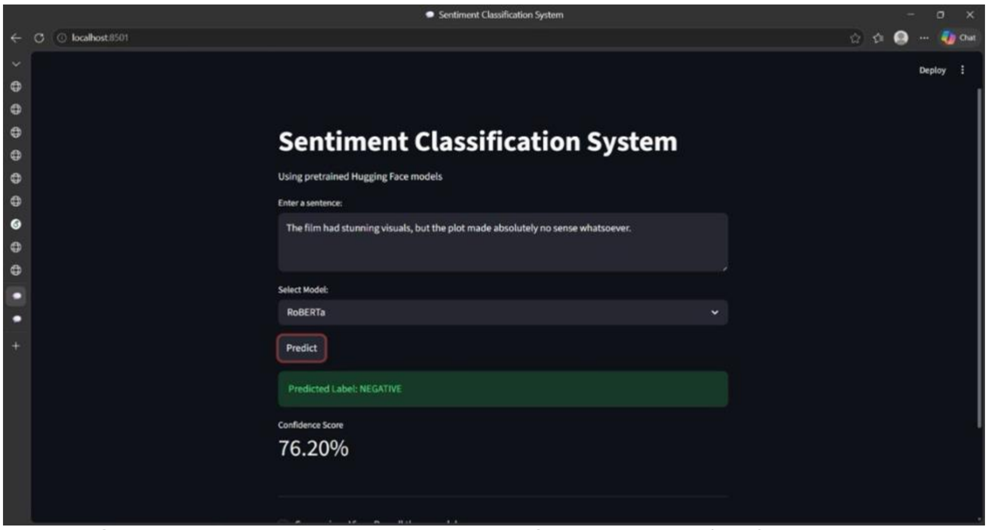

Figure 13: Sentence 3 tested on RoBERTa — Predicted NEGATIVE with 76.20% confidence.

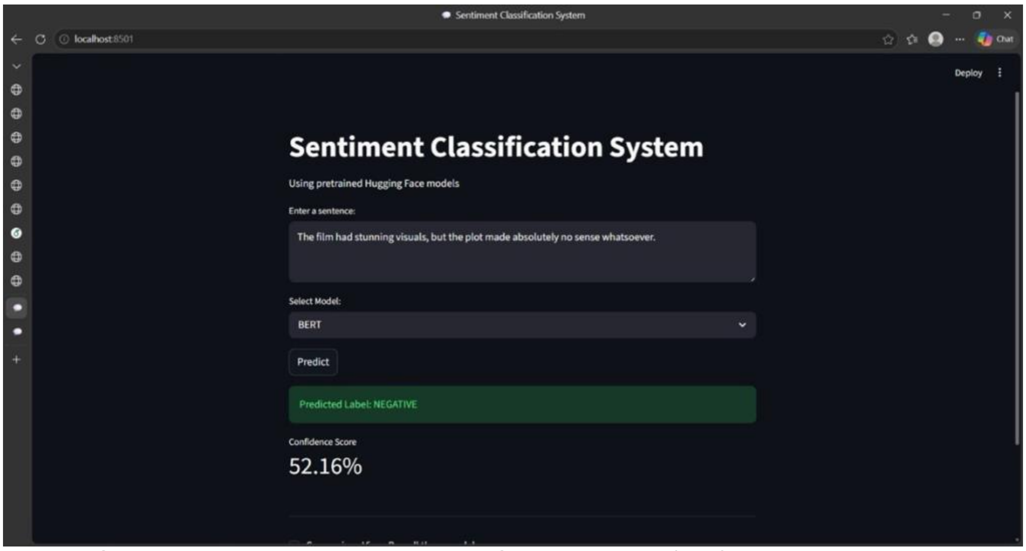

Figure 14: Sentence 3 tested on BERT — Predicted NEGATIVE with 52.16% confidence.

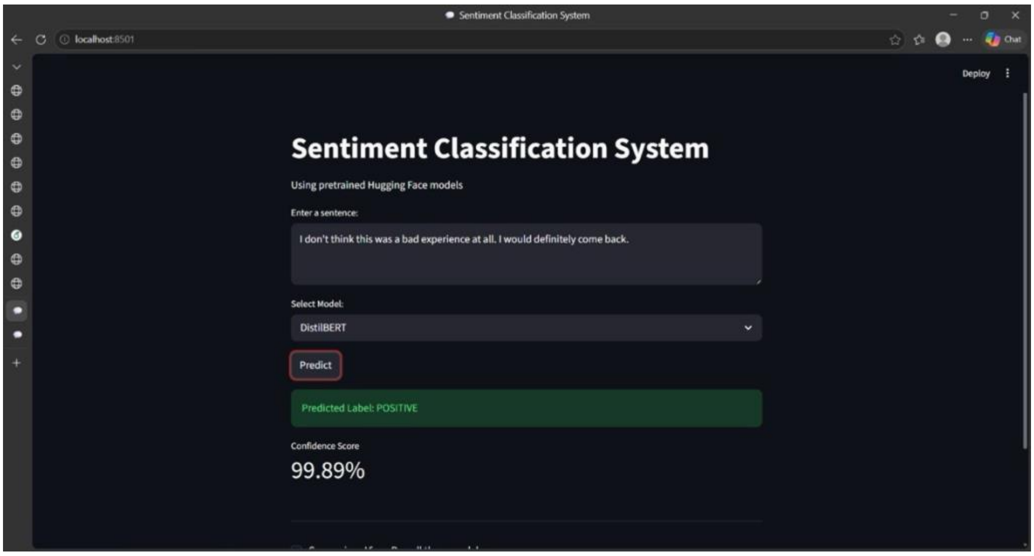

Figure 15: Sentence 4 tested on DistilBERT — Predicted POSITIVE with 99.89% confidence.

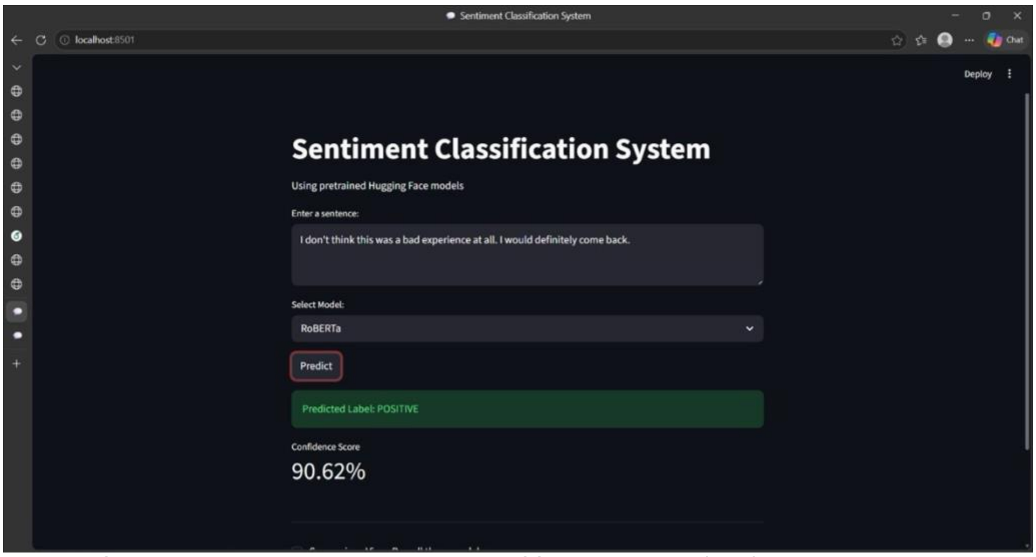

Figure 16: Sentence 4 tested on RoBERTa — Predicted POSITIVE with 90.62% confidence.

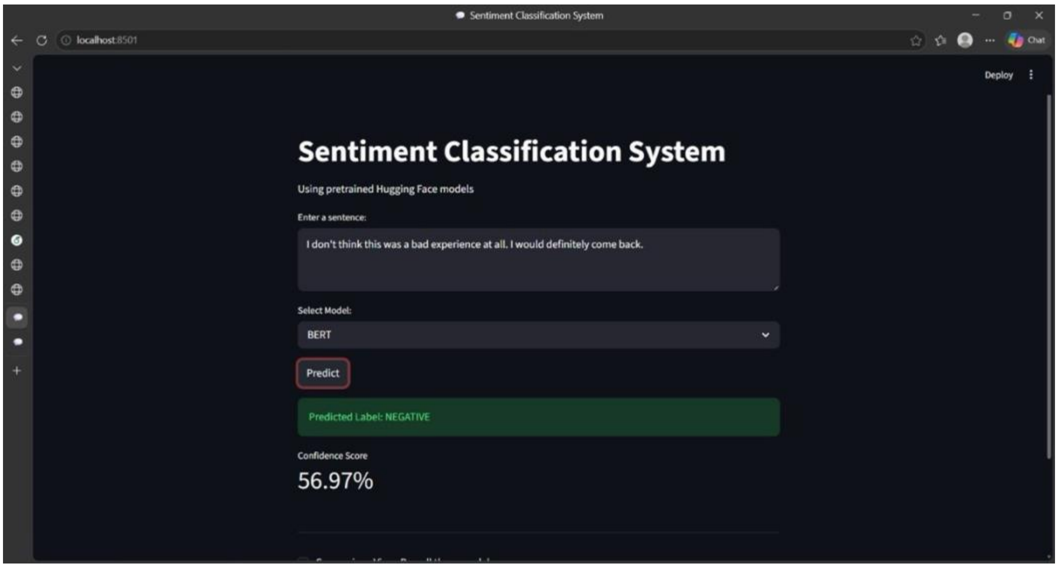

Figure 17: Sentence 4 tested on BERT — Predicted NEGATIVE with 56.97% confidence.

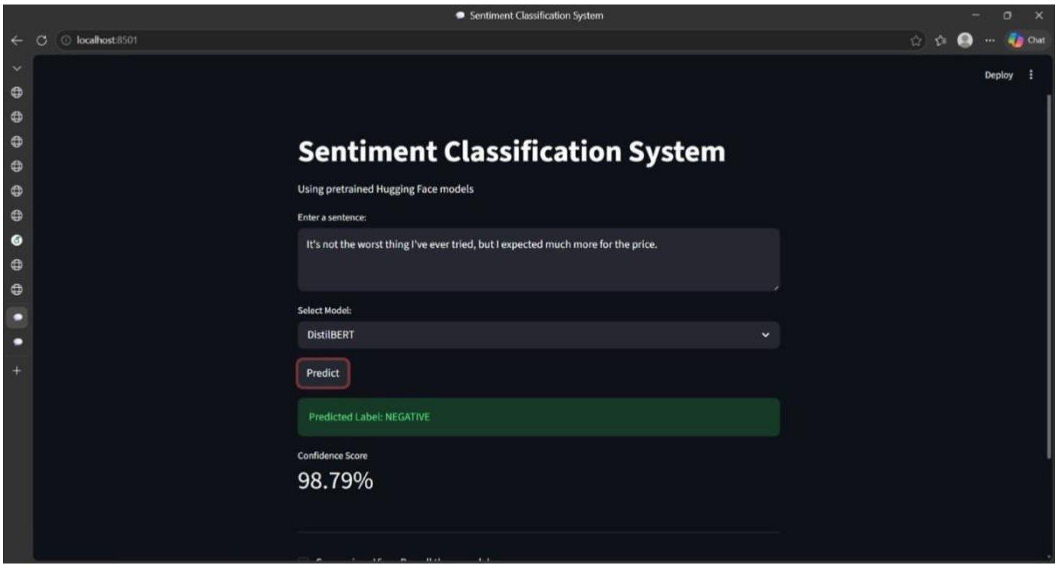

Figure 18: Sentence 5 tested on DistilBERT — Predicted NEGATIVE with 98.79% confidence.

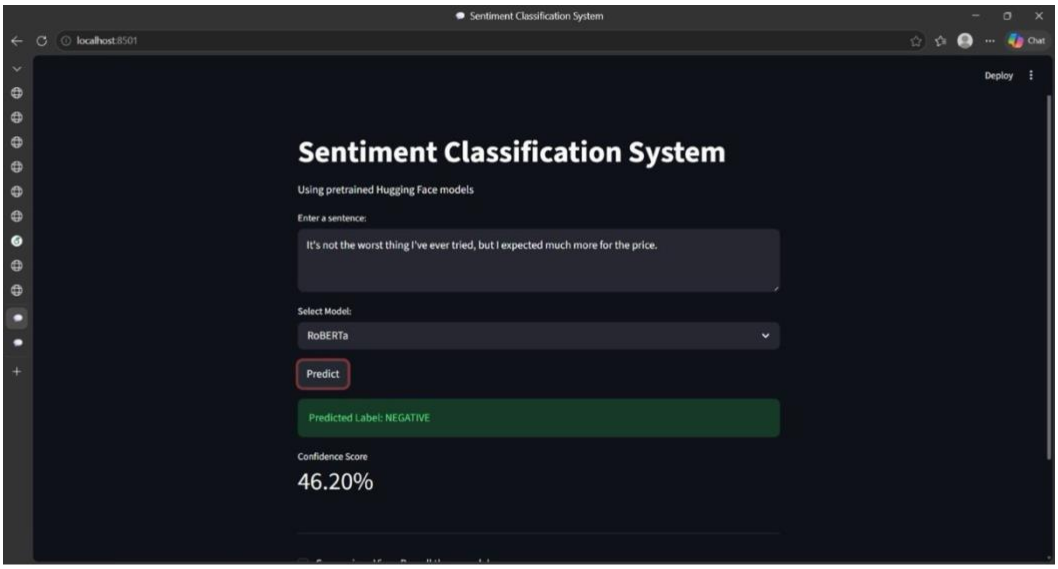

Figure 19: Sentence 5 tested on RoBERTa — Predicted NEGATIVE with 46.20% confidence.

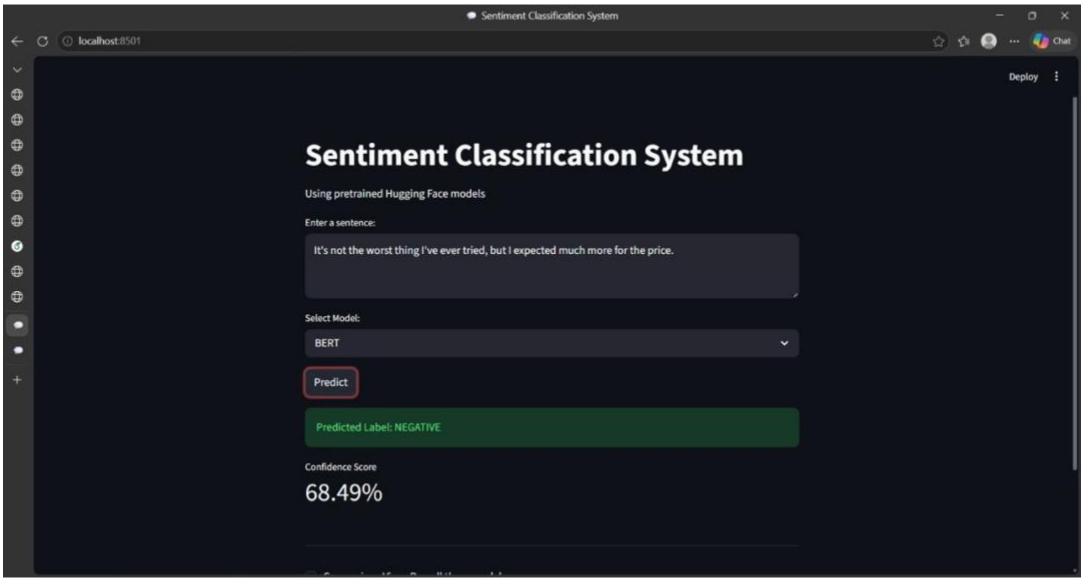

Figure 20: Sentence 5 tested on BERT — Predicted NEGATIVE with 68.49% confidence..

10. Discussion

The experimental results demonstrated that transformer-based models are effective in sentiment analysis tasks. The three models were able to classify text sentiment successfully, but their performance varied in terms of accuracy, confidence scores, and prediction consistency.

According to the evaluation results, DistilBERT achieved the most consistent overall performance among the three models, obtaining the highest F1-Score while maintaining strong confidence scores across the testing sentences. BERT also produced reliable predictions and achieved comparable performance. RoBERTa demonstrated effective sentiment classification capabilities, although its performance was slightly lower on the evaluation metrics used in this project.

The comparison highlighted the strengths and differences between the three models. Overall, transformer-based models proved to be powerful tools for sentiment classification applications and provided valuable insights into the effectiveness of modern NLP techniques.

11. Conclusion

In this project, a sentiment analysis system was successfully developed using three transformer-based models: BERT, RoBERTa, and DistilBERT. The system was implemented using Streamlit to provide an interactive and user-friendly interface for sentiment classification.

The project demonstrated the effectiveness of transformer models in analyzing text sentiment and comparing model performance. Each model showed different strengths in terms of accuracy, confidence scores, and prediction speed.

Overall, the project provided practical experience in Natural Language Processing, transformer models, and web application development using Streamlit.

12. References

1. Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018).
   BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.
   Available at:
   BERT Research Paper

---

2. Liu, Y., Ott, M., Goyal, N., et al. (2019).
   RoBERTa: A Robustly Optimized BERT Pretraining Approach.
   Available at:
   RoBERTa Research Paper

---

3. Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019).
   DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter.
   Available at:
   DistilBERT Research Paper

---

4. Hugging Face Documentation.
   Available at:
   Hugging Face Transformers Documentation

---

5. Streamlit Documentation.
   Available at:
   Streamlit Documentation

---

6. Vaswani, A., Shazeer, N., Parmar, N., et al. (2017).
   Attention Is All You Need.
   Available at:
   Transformer Research Paper